

AI StudyBuddy – Short Summary

1. Project Overview

AI StudyBuddy is an AI-powered learning platform that helps students study efficiently. It uses Node.js, Express.js, MongoDB, JWT authentication, bcrypt, and Google Gemini AI.

2. Problem

Students often have lengthy and scattered study materials. Manually creating notes, flashcards, quizzes, and study schedules takes time and makes revision difficult.

3. Solution

The system provides one centralized platform where authenticated users can upload study material and automatically generate summaries, flashcards, multiple-choice quizzes, and personalized study plans using Gemini AI.

4. Main Features

- JWT-based secure login and authentication
- Study material upload and management
- AI-generated concise summaries
- AI-generated flashcards
- AI-generated MCQ quizzes
- Personalized study plans
- Student/Admin role-based access
- Storage of generated resources in MongoDB

5. Technology Stack

- Frontend: React.js
- Backend: Node.js + Express.js
- Database: MongoDB + Mongoose
- AI: Google Gemini API
- Security: JWT + bcryptjs
- Testing: Postman / ThunderClient

6. Architecture

The backend follows a modular RESTful and MVC architecture. Main layers include authentication middleware, route handlers, controllers, AI service, database/models, and API responses. This improves maintainability, debugging, and scalability.

7. Database Entities

- User – name, email, hashed password, role
- Study Material – title, subject, content, user reference
- Summary – AI-generated summary
- Flashcard – question and answer
- Quiz – generated questions
- Study Plan – plan and exam date

8. User Flow

User registers/logs in → JWT authentication → uploads study material → selects an AI feature → Gemini processes the content → summary/flashcards/quiz/study plan is generated → resource is displayed and stored for future use.

9. Important API Endpoints

- POST /api/auth/register – User registration
- POST /api/auth/login – User login
- POST /api/material/upload – Upload study material
- POST /api/materials/:id/summarize – Generate summary
- POST /api/ai/flashcards – Generate flashcards
- POST /api/ai/quiz – Generate quiz
- POST /api/ai/study-plan – Generate study plan

10. Requirements

Software: Windows/macOS/Linux, Node.js v16+, npm v8+, Express.js, MongoDB, Postman, VS Code. Hardware: Intel i5 8th Gen/AMD Ryzen 5 or equivalent, minimum 8 GB RAM, and about 1 GB available storage.

11. Outcome

AI StudyBuddy reduces manual study preparation time and supports faster revision and personalized learning. Its modular design also allows future AI features to be added with minimal architectural changes.